

TASK – 5

Exploratory Data Analysis (EDA) Report – Titanic Dataset

1. Introduction:

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset through statistical summaries and visualizations. The objective of this project is to analyze the Titanic dataset, identify patterns, detect missing values and outliers, and understand the factors that influenced passenger survival.

2. Dataset Overview:

The Titanic dataset contains information about passengers who traveled on the Titanic ship. The dataset includes demographic details, ticket information, passenger class, fare, and survival status.

Dataset Details

- Total Records: 891
- Total Features: 12
- Target Variable: Survived
- Data Source: Titanic Training Dataset

Important Features

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

3. Data Cleaning Check :

The dataset was inspected for data quality issues before analysis.

Missing Values

- Age column contains missing values.
- Cabin column contains a large number of missing values.
- Embarked column contains a few missing values.

Duplicate Records

- No significant duplicate records were found.

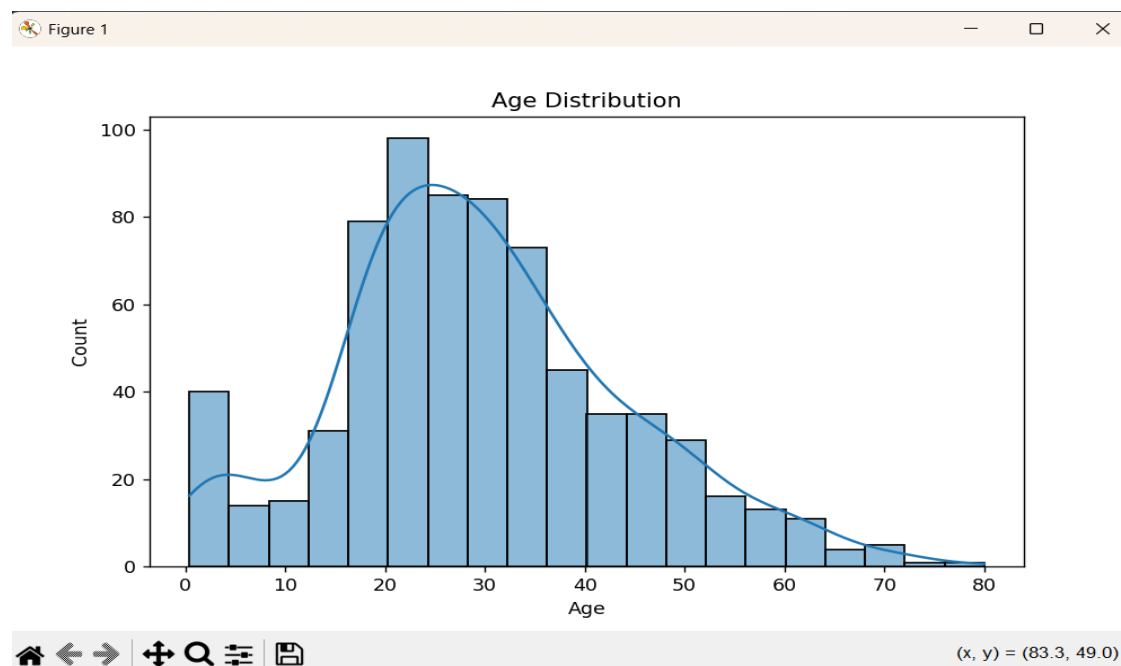
Data Types

- Numerical Features: Age, Fare, SibSp, Parch, Pclass
- Categorical Features: Sex, Embarked, Cabin

Outlier Detection

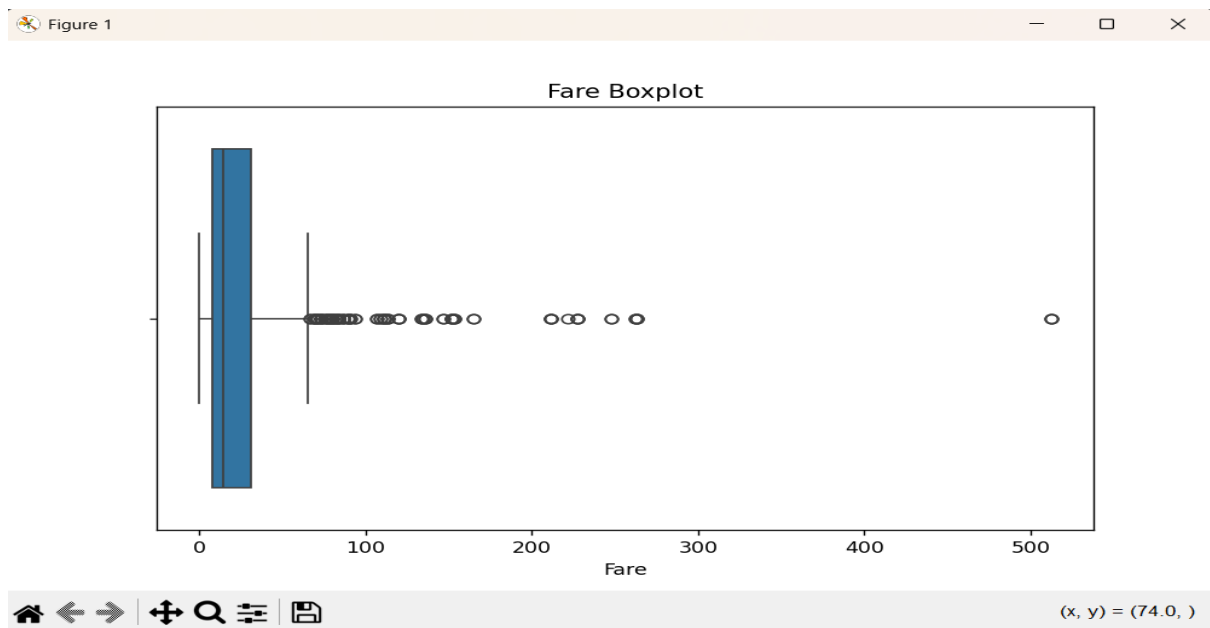
Boxplot analysis revealed outliers in the Fare column, indicating some passengers paid exceptionally high ticket prices.

4. Visualizations :



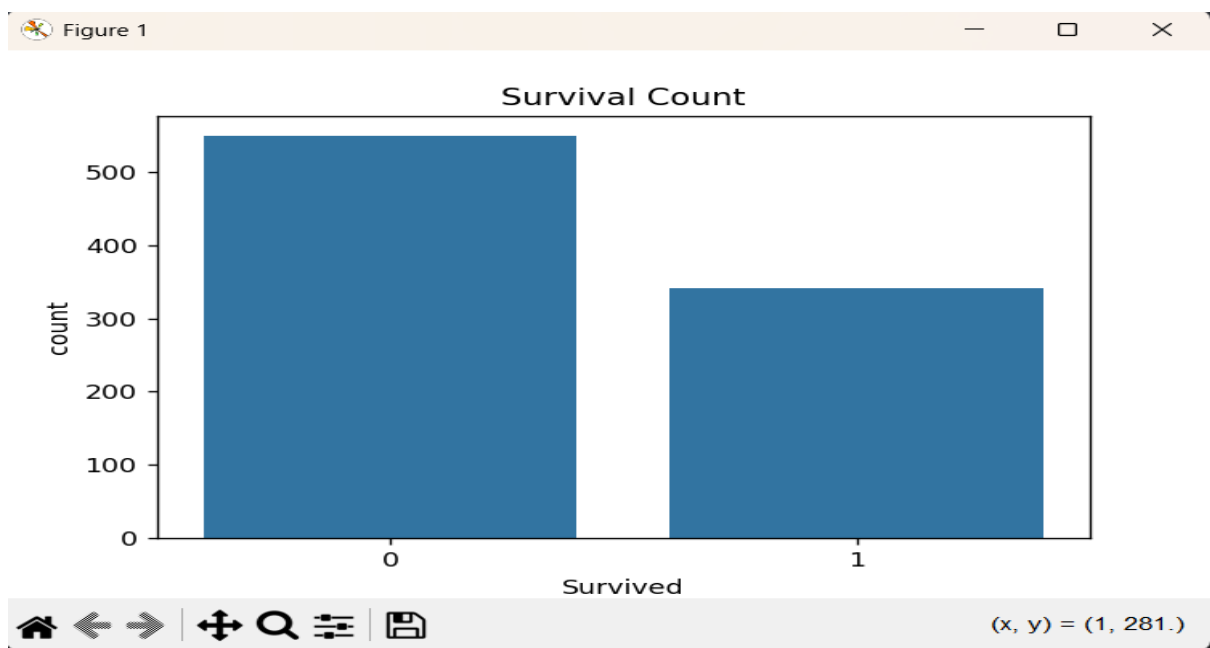
4.1 Age Distribution

Histogram used to understand the age distribution of passengers.



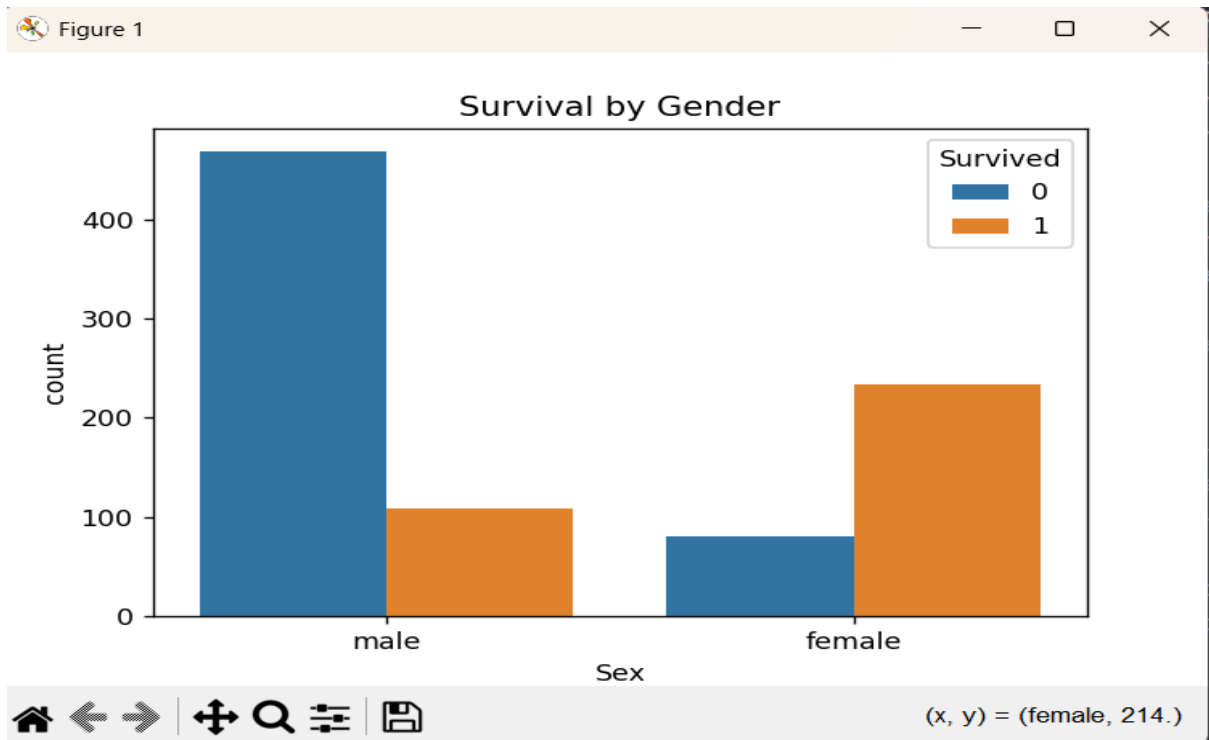
4.2 Fare Boxplot

Boxplot used to identify fare spread and outliers.



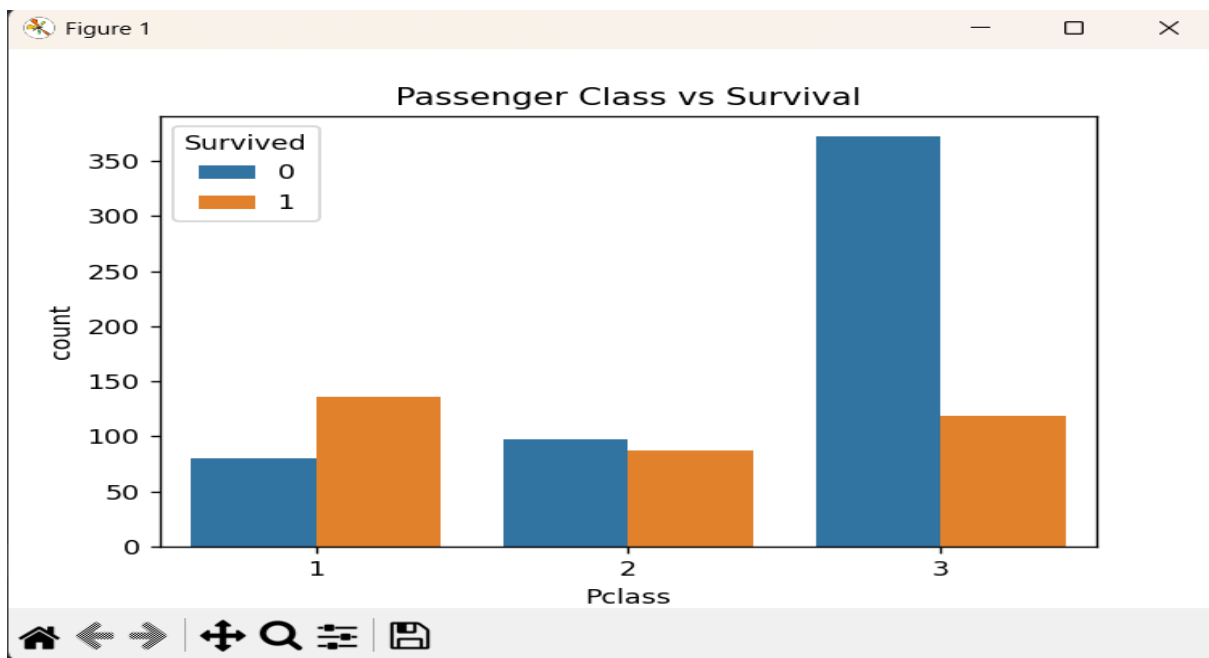
4.3 Survival Count

Count plot showing the number of survivors and non-survivors.



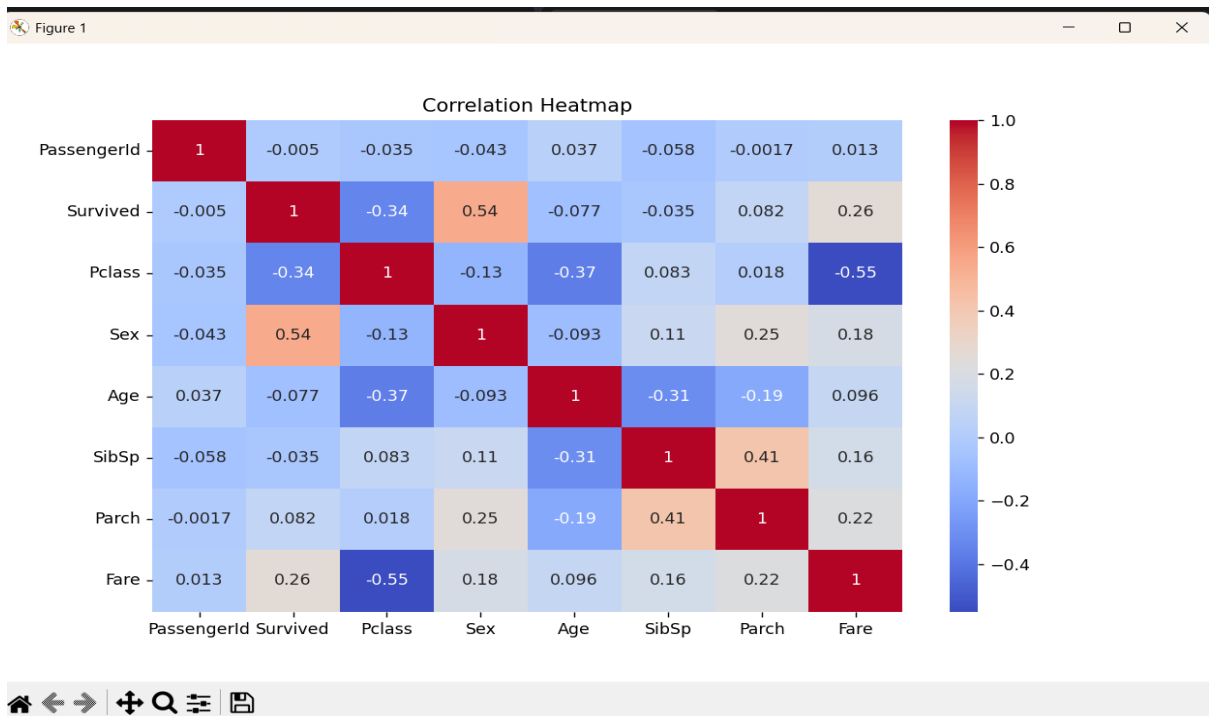
4.4 Survival by Gender

Comparison of survival rates between male and female passengers.



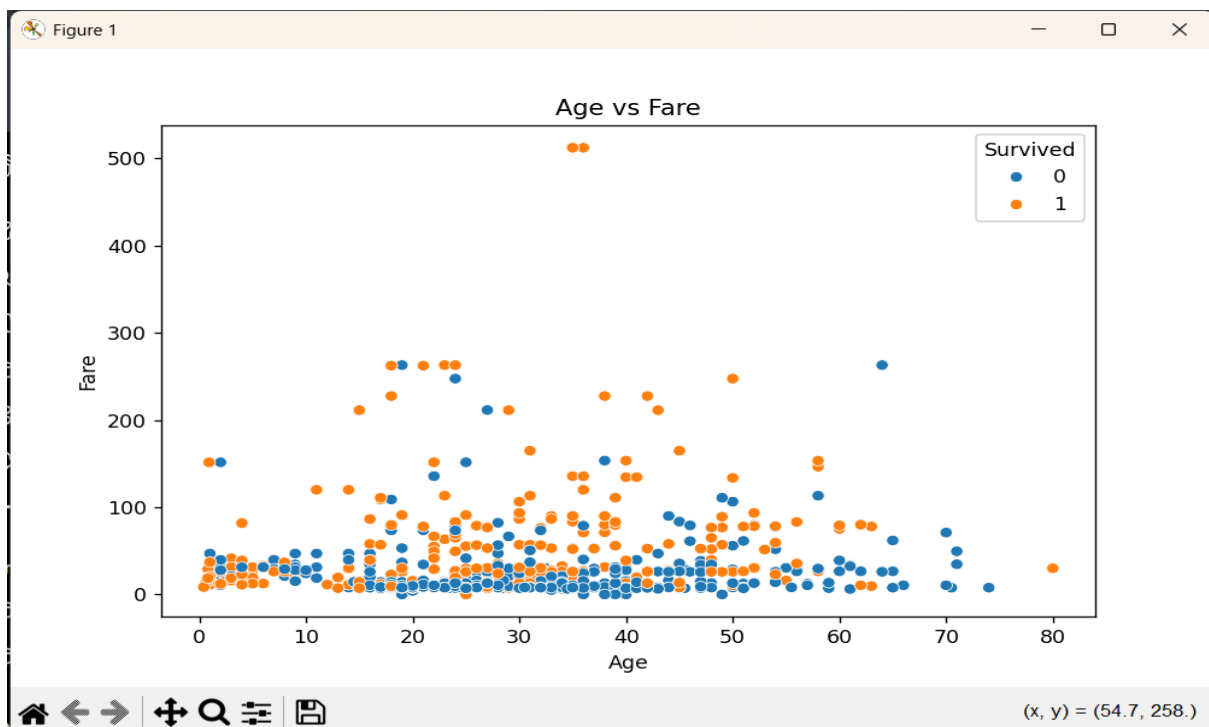
4.5 Passenger Class vs Survival

Analysis of survival based on passenger class (1st, 2nd, and 3rd).



4.6 Correlation Heatmap

Heatmap displaying relationships among numerical features.



4.7 Age vs Fare Scatter Plot

Scatter plot showing the relationship between passenger age and ticket fare.

5. Observations:

- ☐ Most passengers were between 20 and 40 years old.
- ☐ Female passengers had a significantly higher survival rate than male passengers.
- ☐ First-class passengers were more likely to survive than passengers in lower classes.
- ☐ Fare contains several extreme values (outliers).
- ☐ Passenger class and gender strongly influenced survival.
- ☐ Age showed a weaker correlation with survival compared to other features.

6. Summary Findings:

- ☐ Gender is one of the strongest predictors of survival.
- ☐ Passenger class has a significant impact on survival probability.
- ☐ Higher fare-paying passengers generally had better survival chances.
- ☐ Missing values are primarily concentrated in the Age and Cabin columns.
- ☐ Visualizations effectively highlighted trends and relationships within the dataset.

7. Conclusion:

The Exploratory Data Analysis of the Titanic dataset provided valuable insights into passenger survival patterns. Gender, passenger class, and fare emerged as the most influential factors affecting survival. The analysis also identified missing values and outliers that should be addressed before building predictive machine learning models. This EDA serves as a strong foundation for future classification and survival prediction tasks.